LESSON 3.2.2

Encryption of AI Data

This lesson explores how encryption protects AI data confidentiality and integrity across the three states of data: **at rest**, **in transit**, and **in use** including modern solutions like TEE, Homomorphic Encryption, and DP-SGD.

What Is Data Encryption?

Data encryption transforms readable data into an unreadable format. Only systems or users with valid **decryption keys** can access the original data.

Confidentiality

Prevents unauthorized access to sensitive data

Integrity

Ensures data has not been tampered with

DATA AT REST

Protection Methods

Data at rest is stored data that is not actively being accessed or transmitted. It is protected through encryption applied at multiple levels:

1

File-Level Encryption

Encrypts individual files directly on the storage medium

2

Folder-Level Encryption

Applies encryption to entire directories for grouped protection

3

Storage-Drive Encryption

Full-disk tools like BitLocker encrypt all data on the drive

DATA AT REST

Key Storage Options

Encryption is only as strong as its key management. Keys for data at rest can be stored and managed using specialized hardware and cloud services:

KMS

Key Management Service — controls access via access control policies

TPM

Trusted Platform Module — hardware-based, tamper-resistant key storage

HSM

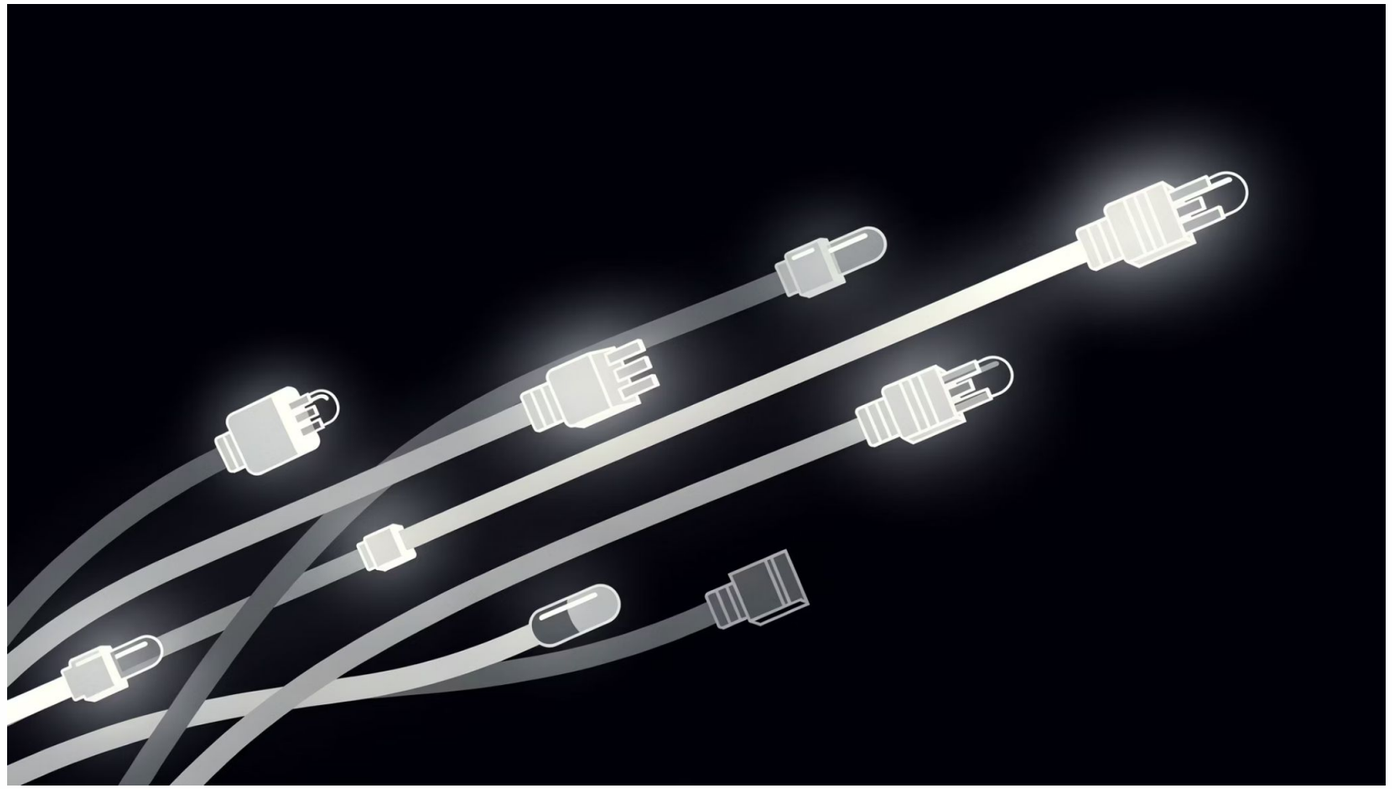
Hardware Security Module — dedicated device for cryptographic operations

Cloud KMS

AWS KMS & Azure Key Vault — integrate natively with cloud AI deployments

DATA IN TRANSIT

Protecting Data in Motion



Data in transit moves between locations — networks, systems, or cloud endpoints. Standard protocols provide encryption during transmission:

TLS 1.3+

Transport Layer Security — industry standard; version 1.3 or newer is recommended

VPN

Virtual Private Network — creates an encrypted tunnel over public networks

Secure Protocols

SFTP, FTPS, and HTTPS for encrypted file and web transfers

DATA IN TRANSIT

HTTP-3 and QUIC

What Is QUIC?

Quick UDP Internet Connections (QUIC) is a modern transport protocol developed by Google. HTTP-3 is built on top of QUIC, replacing TCP with UDP for faster, lower-latency communication.

Key Advantage: QUIC natively bundles TLS 1.3, meaning encryption is built into the connection — not layered on top. This reduces handshake overhead and eliminates a class of downgrade attacks.

Faster

UDP-based transport reduces latency vs. TCP

Secure by Default

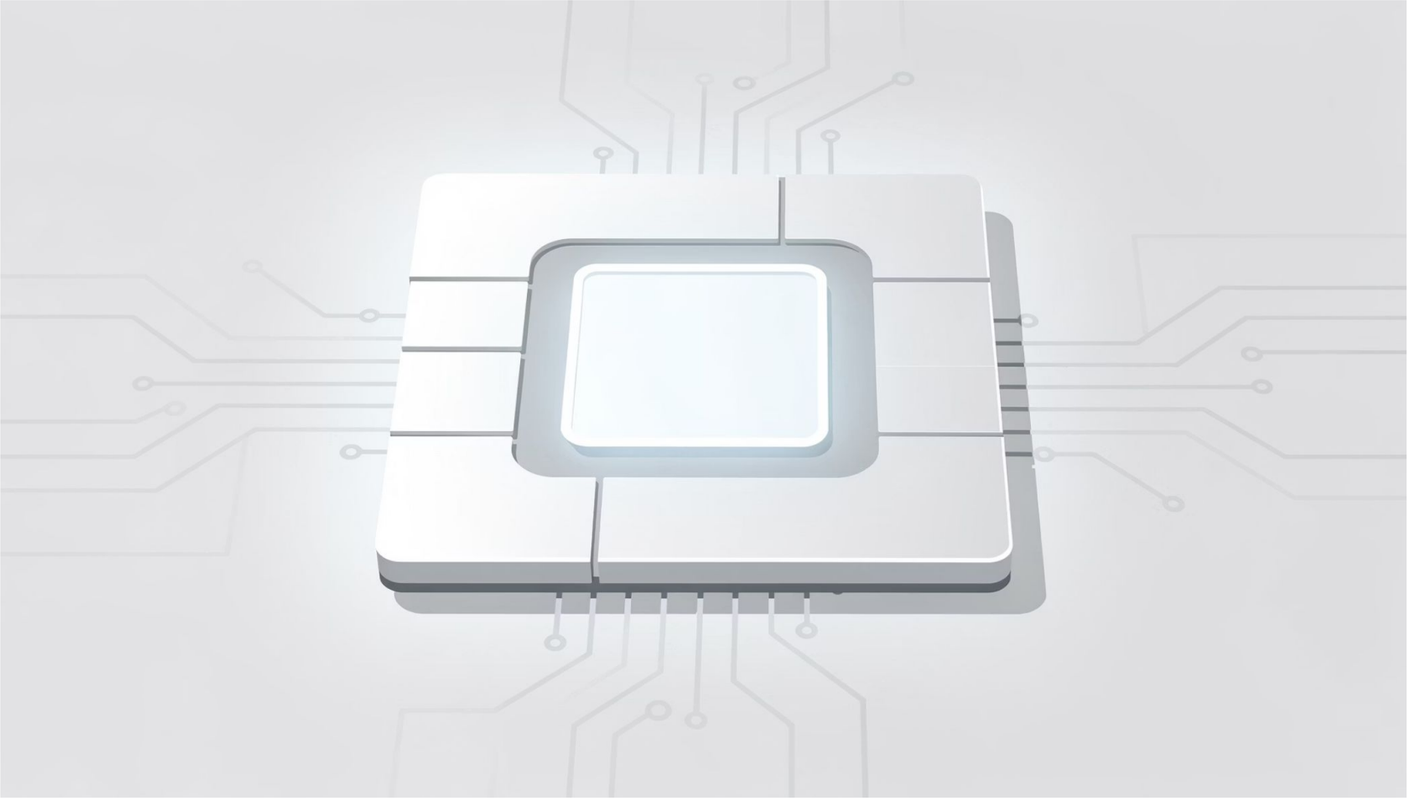
TLS 1.3 is natively included — not optional

Modern Standard

Recommended for AI systems with high-throughput data pipelines

DATA IN USE

Trusted Execution Environments (TEE)



A **Trusted Execution Environment (TEE)** is a hardware-isolated region of a processor that ensures sensitive data is processed in a protected enclave — inaccessible to the host OS or hypervisor.

AWS Nitro Enclaves

Isolated compute environments for sensitive cloud workloads

Azure Confidential Computing

Hardware-backed enclaves for AI inference and training

DATA IN USE

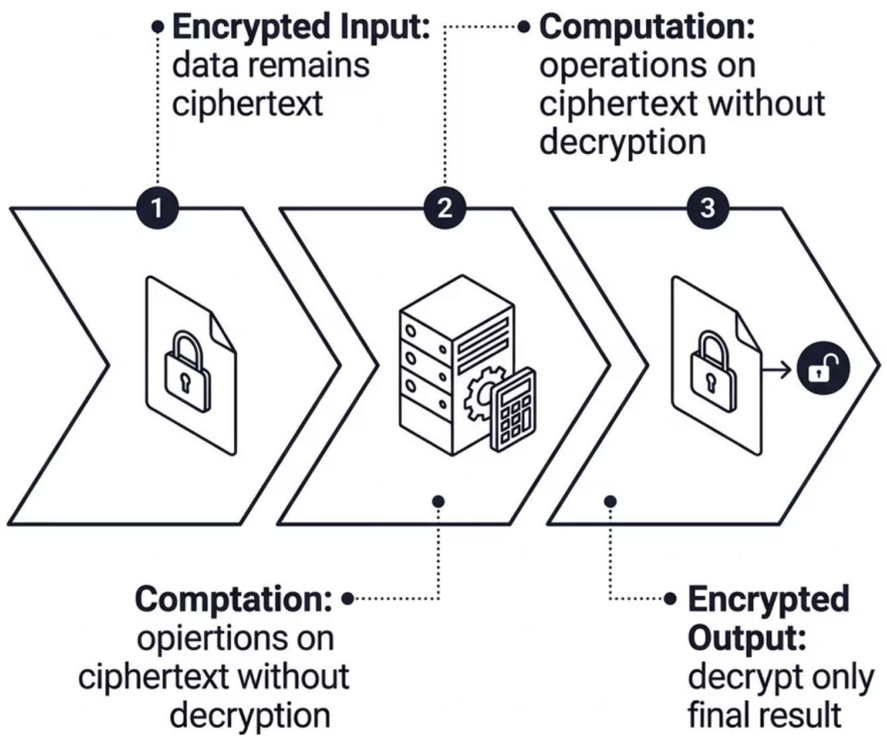
Homomorphic Encryption (HE)

How It Works

Homomorphic Encryption (HE) allows mathematical operations to be performed directly on ciphertext without ever decrypting the data. The result, when decrypted, matches the output of the same operation on plaintext.

This eliminates the exposure window entirely: data stays encrypted from storage through computation to output.

> **AI Use Case:** HE enables secure AI inference on sensitive data — the model computes predictions on encrypted inputs without accessing raw values.

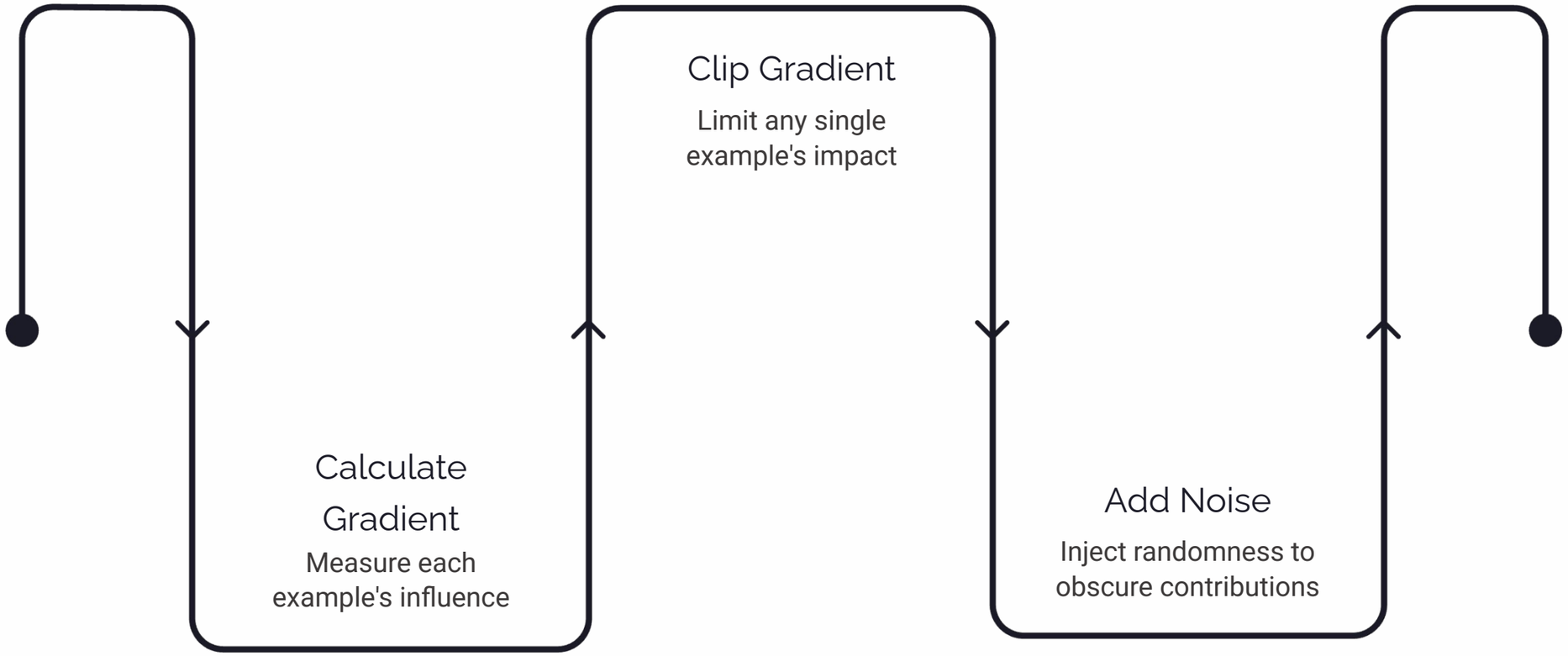


MACHINE LEARNING

DP-SGD: Privacy in ML

Differentially Private Stochastic Gradient Descent (DP-SGD) protects individual data points used during model training preventing an attacker from inferring whether specific records were included.

Training



More Noise

Stronger privacy protection against inference attacks

Trade-Off

Higher noise degrades model accuracy — balance must be tuned

Key Takeaways

1

Encryption Fundamentals

Encryption makes data unreadable without valid decryption keys, protecting both confidentiality and integrity.

2

Data at Rest

Protected via file, folder, or drive encryption. Keys stored in KMS, TPM, HSM, or cloud services like AWS KMS and Azure Key Vault.

3

Data in Transit

Secured with TLS 1.3+, VPNs, and protocols (SFTP, FTPS, HTTPS). HTTP-3/QUIC offers modern, built-in TLS 1.3 protection.

4

Data in Use

Hardest to protect. TEE (secure enclaves) uses hardware isolation; Homomorphic Encryption enables computation on encrypted data.

5

DP-SGD

Protects ML training privacy via gradient clipping and noise injection. Trade-off: more noise = stronger privacy, lower accuracy.